Integer STU flux families: positive masses and tadpole-controlled coupling

29 October 2026

Abstract

An exact even integral STU flux lattice admits a positive-divisor family of no-scale complex-structure/axiodilaton critical points. With $A_I B_I = k$, its charge is $Q = 8k$, its full real F-term Jacobian has determinant $-4096k^4$, and its eight canonical masses squared are k times a fixed positive spectrum. An integral modular image gives $S = \mathrm{i}k^2$, $U_i = \mathrm{i}k$, hence $g_s = k^{-2}$. At inherited quotient tadpole capacity $L = 28$, the construction reaches $g_s = 1/9$ with $Q = 24$ and contains nine distinct reduced moduli point types. We also prove exact pointwise charge minima at those points using primitive complementary-monomial pairs and a scaling obstruction. The results are theorems of the stated period/lattice model. Its global orientifold interpretation is inherited, and the nonzero superpotential, unstabilized Kähler sector and common volume factor remain explicit.

1 The exact model and its interpretation

Flux stabilization and its source-charge budget are related but distinct parts of the no-scale compactification problem [1–3]. Here we solve a specified low-dimensional integral period problem; no general tadpole conjecture is proved or refuted. Write $z = (S, U_1, U_2, U_3)$, with $y_A = \operatorname{Im} z_A > 0$, and take

$$\Pi = (1, U_1, U_2, U_3, U_1U_2U_3, -U_2U_3, -U_1U_3, -U_1U_2)^T, \tag{1}$$

$$P(n) = n_0 + \sum_i n_i U_i - n^0 U_1U_2U_3 + n^1 U_2U_3 + n^2 U_1U_3 + n^3 U_1U_2, \tag{2}$$

$$W = P(f) - SP(h), \qquad K = -\sum_{A=0}^{3} \log(2y_A). \tag{3}$$

The flux ordering is $(f^I, f_I; h^I, h_I)$, $I = 0, 1, 2, 3$, with $(f, h) \in (2\mathbb{Z})^{16}$. Fix the quotient normalization

$$Q = \frac{1}{2}\sum_I (f_I h^I - f^I h_I), \qquad N_{\mathrm{cover}} = 2Q. \tag{4}$$

All results below use this precise sign and factor of two.

The geometric motivation is the exact untwisted three-complex-structure period sector of the resolved toroidal model associated with Denef et al. [2]. The smooth orientifold realization, the interpretation of this even sublattice as a conservative non-exotic O3-minus sector, and the source budgets $L = 28$ and $L = 4$ are inherited from the preceding source calculations. This note does not independently establish those global geometric or admissibility assertions. Its algebraic claims hold for (1)–(4) whether or not one inserts either capacity.

An F-term zero means $D_A W = 0$ for these four complex variables. We use the eight-by-eight real Jacobian J_F of $(\Re D_A W, \Im D_A W)$, in interleaved real coordinates $(\Re z_0, \Im z_0, \ldots, \Re z_3, \Im z_3)$. This is the rank relevant to the no-scale potential; a holomorphic Hessian alone is insufficient when $W \neq 0$.

2 An integral divisor family

Theorem 1 (Construction and masses)**.** *Let $k > 0$ and A_I, B_I be positive integers satisfying $A_I B_I = k$ for every I. Set $f^I = h_I = 0$, $f_I = -2A_I$, $h^I = -2B_I$. Then*

$$U_i = \mathrm{i}\frac{A_0}{A_i}, \qquad S = \mathrm{i}\frac{A_1 A_2 A_3}{A_0 k} \tag{5}$$

is an F-term zero with $Q = 8k$ and $\det J_F = -4096k^4$. With the common Kähler-sector factor e^{K_K} set to one, its canonical masses squared are

$$k\{28 - 12\sqrt{5},\ 28 + 12\sqrt{5},\ (12 - 4\sqrt{5})^{[3]},\ (12 + 4\sqrt{5})^{[3]}\}. \tag{6}$$

The superscript [3] denotes multiplicity three. All masses squared are strictly positive.

Proof. The unit seed has $f_I = h^I = -2$, other entries zero, and

$$W_0 = -2(1 + U_1 + U_2 + U_3) - 2S(U_1U_2U_3 - U_1U_2 - U_1U_3 - U_2U_3). \tag{7}$$

At $z_A = \mathrm{i}$, $W_0 = -4 - 12\mathrm{i}$ and every holomorphic first derivative equals $-6 + 2\mathrm{i}$. Since $K_A = \mathrm{i}/2$ there, all four F-terms vanish. Direct real differentiation gives the seed Jacobian in Appendix A, with determinant -4096.

For $c = A_0$ and the positive imaginary parts y_A in (5), coefficient matching gives

$$W(z) = cW_0(z_0/y_0, z_1/y_1, z_2/y_2, z_3/y_3), \qquad \prod_A y_A = A_0^2/k. \tag{8}$$

The logarithmic Kähler derivatives imply $D_A W = (c/y_A)(D_A W_0)(z/y)$. Thus the rescaled seed is a zero. The divisor conditions ensure that all flux entries were integers before performing this construction; no continuous flux is rounded. Equation (4) gives $Q = \frac{1}{2}\sum_I 4A_I B_I = 8k$.

Rows and columns of the real Jacobian scale separately, so

$$\det J_F = \frac{c^8}{(\prod_A y_A)^4} \det J_{F,0} = -4096k^4. \tag{9}$$

At an F-term zero the real potential Hessian and kinetic metric are

$$H_{\mathrm{real}} = 2e^K J_F^T \,\mathrm{diag}(4y_0^2, 4y_0^2, \ldots, 4y_3^2, 4y_3^2) J_F, \tag{10}$$

$$G_{\mathrm{real}} = \mathrm{diag}(1/(2y_0^2), 1/(2y_0^2), \ldots, 1/(2y_3^2), 1/(2y_3^2)). \tag{11}$$

Here $\mathcal{L}_{\mathrm{kin}} = -\frac{1}{2}G_{ab}\partial\phi^a\partial\phi^b$. Canonical normalization of (10) and (8) gives

$$H_{\mathrm{can}} = \frac{c^2}{\prod_A y_A} H_0 = kH_0 \tag{12}$$

in aligned frames. At the unit seed, $H_0 = J_{F,0}^T J_{F,0}$ and

$$\det(tI - H_0) = (t^2 - 56t + 64)(t^2 - 24t + 64)^3. \tag{13}$$

Its roots give (6). Positivity follows, for example, from $28^2 > 144 \cdot 5$ and $12^2 > 16 \cdot 5$. □

The simple subfamily $A_I = m$, $B_I = n$, $k = mn$ gives $S = \mathrm{i}m/n$, $U_i = \mathrm{i}$, $Q = 8mn$. A common integer multiple of (m, n) changes the charge while leaving the moduli point fixed; it is not a new moduli point. More generally (8) is a map between different integral models after the divisor condition is imposed. It is not a continuous family of fixed-flux isolated vacua or an assumed integral duality.

k	$(\operatorname{Im} S; \operatorname{Im} U_1, \operatorname{Im} U_2, \operatorname{Im} U_3)$	$Q_{\min}$ at each point
1	$(1;1,1,1)$	8
2	$(2;1,1,1)$, $(1;1,1,2)$	16
	$(2;1,2,2)$, $(4;2,2,2)$	16
3	$(3;1,1,1)$, $(1;1,1,3)$	24
	$(3;1,3,3)$, $(9;3,3,3)$	24

Table 1: Nine reduced moduli point types attained within capacity 28. The minima are over the full stated even lattice at each point, before imposing capacity. They are not total landscape counts.

3 A weak-coupling branch and modular point types

Choosing $A_0 = 1$, $A_i = k$ in Theorem 1, then inverting all three torus moduli, gives the explicit even flux

$$\begin{aligned} f^I &= (2, 2k, 2k, 2k), \quad f_I = h^I = (0, 0, 0, 0), \\ h_I &= (-2k, -2, -2, -2), \qquad S = \mathrm{i}k^2, \quad U_i = \mathrm{i}k. \end{aligned} \tag{14}$$

It can also be checked directly in

$$W = -2U_1U_2U_3 + 2k\sum_{i<j} U_iU_j + 2S(k + U_1 + U_2 + U_3). \tag{15}$$

At (14),

$$W = 4k^3(-3 + \mathrm{i}), \qquad Q = 8k, \qquad g_s = k^{-2}. \tag{16}$$

The determinant and spectrum remain those in Theorem 1. Capacity imposes $k \le \lfloor L/8 \rfloor$ on this branch. Thus $L = 28$ permits $k = 1, 2, 3$, with respective couplings $1, 1/4, 1/9$ and residual quotient D3 charges $20, 12, 4$. At $k = 3$ the flux is $(2, 6, 6, 6; 0^4; 0^4; -6, -2, -2, -2)$, $N_{\text{cover}} = 48$, and $Q = 24$. Comparing 48 directly to capacity 28 would use inconsistent normalizations.

Along this branch $g_s = (8/Q)^2$. The weaker coupling at finite allowed charge is a useful construction, but a parametrically weak-coupling limit requires increasing capacity. Within the whole positive divisor family, $k^{-2} \le \operatorname{Im} S \le k^2$, so modular reduction cannot improve on $g_s = 1/k^2$ in that branch. Optimality over other even integral fluxes is not established. For $L \ge 8$, the attainable branch minimum is precisely $g_s = \lfloor L/8 \rfloor^{-2}$, realized by its largest allowed integer k.

For clarity, separate new moduli points from their modular images. With $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in SL(2, \mathbb{Z})$,

$$S' = \frac{aS + b}{cS + d}, \quad f' = af + bh, \quad h' = cf + dh, \quad W' = \frac{W}{cS + d}. \tag{17}$$

These images preserve Q, even integrality, F-term zeros and masses, and are not counted again. Torus inversion is the actual integer polynomial map $W'(U_j') = U_j' W(-1/U_j')$; torus permutations relabel the U_i.

For $k = 1, 2, 3$ there are $1 + 16 + 16 = 33$ divisor assignments. Reducing each pure imaginary modulus by inversion and sorting the three torus factors gives the nine point types in Table 1. A pure imaginary value $\mathrm{i}y$, $y \ge 1$, is in the usual modular fundamental domain and has a unique such imaginary representative. The listed tuples are therefore distinct under separate modular factors and torus permutations. This does not classify all flux multiplicities at a fixed point or extra string dualities mixing S and U.

4 Exact pointwise minima

Let C_e be the real coefficients of the 16 multiaffine monomials of W, with exponent vectors $e \in \{0,1\}^4$. The map from flux to C is a signed permutation, so $C \in (2\mathbb{Z})^{16}$. Define

$$\mathcal{A}(x,y) = \frac{1}{y}\begin{pmatrix} 1 & x \\ x & x^2+y^2 \end{pmatrix}, \quad \mathcal{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{18}$$

Direct tensor algebra gives the normalized positive identity

$$E = \tfrac{1}{4} C^T \bigotimes_A \mathcal{A}(\Re z_A, y_A) C, \quad Q = \tfrac{1}{4} C^T \mathcal{J}^{\otimes 4} C, \tag{19}$$

$$E - Q = e^K \sum_A 4 y_A^2 |D_A W|^2. \tag{20}$$

One derivation uses $u = (1,z)^T$ and $uu^\dagger + \bar{u}u^T = 2y\mathcal{A}$, $uu^\dagger - \bar{u}u^T = 2\mathrm{i}y\mathcal{J}$. The total and alternating sums of $|W|^2$ over the 16 patterns of conjugating individual variables produce the two quadratic forms in (20). Because C is real, three-variable flips are conjugate to complementary one-variable flips; finally $2y_A D_A W = \mathrm{i}W(z_A \mapsto \bar{z}_A)$ establishes the last equality.

At a pure imaginary point, let $\bar{e} = \mathbf{1} - e$ and $w_e = \prod_A y_A^{2e_A - 1}$. The contribution of each complementary pair to $E - Q$ is

$$\frac{1}{4}\left(\sqrt{w_e} C_e - epsilon_e C_{\bar{e}} / \sqrt{w_e}\right)^2, \qquad \epsilon_e = (-1)^{\sum_A e_A}. \tag{21}$$

For $w_e = u/v > 0$ in lowest terms, an F-term zero must therefore have

$$(C_e, C_{\bar{e}}) = 2r(v, \epsilon_e u), \quad r \in \mathbb{Z}, \qquad Q_e = 2uvr^2. \tag{22}$$

This is a saturated pairwise integral parametrization, not a flux scan.

Lemma 1 (Necessary number of complementary pairs)**.** *At a pure imaginary F-term zero, full real rank requires at least four nonzero complementary monomial pairs in W.*

Proof. If there are $r \le 3$ pairs, choose real scaling weights ℓ_A satisfying $(2e_j - \mathbf{1}) \cdot \ell = 0$ for each supported pair. There are at least $4 - r$ independent choices. Every supported monomial has weight $w = \frac{1}{2}\sum_A \ell_A$, so under $z_A \mapsto e^{t\ell_A} z_A$,

$$W \mapsto e^{tw} W, \qquad D_A W \mapsto e^{t(w-\ell_A)} D_A W. \tag{23}$$

The upper half-planes are preserved and the same fixed-flux zero has a continuous orbit. Its independent tangent vectors $\delta z_A = \ell_A z_A$ lie in the real Jacobian kernel. Discrete modular identifications do not remove these infinitesimal directions. □

Proposition 1. *For every integer $k > 1$, the four point types*

$$(k;1,1,1), \quad (1;1,1,k), \quad (k;1,k,k), \quad (k^2;k,k,k) \tag{24}$$

have exact pointwise minimum $Q_{\min} = 8k$ in the stated even lattice. At $(1;1,1,1)$ the minimum is eight.

Proof. At the unit point each primitive nonzero pair in (22) costs two. At any of the four other types, w_e is an odd integral power of k; hence $2uv \ge 2k$. Four supported pairs therefore cost at least $8k$. For example at $(k^2;k,k,k)$ their primitive costs are $2k^5$ once, $2k^3$ three times and $2k$ four times. The divisor witnesses and their integral modular images attain the bound with full rank, proving equality. □

The positivity and integral-pair analysis apply to the entire even lattice at each specified point, not only to the positive divisor branch. They do not give a global lower bound over all moduli. In particular capacity four fails at these points, but a global no-vacuum theorem at capacity four is not inferred.

5 Scope and reproducibility

All masses in (6) acquire the common positive factor e^{K_K} when the unstabilized Kähler sector is restored. In the displayed normalization $m_{3/2}^2 = 10k$; the mass ratios are fixed, with no claimed tunable hierarchy. Since $W \neq 0$, these are no-scale CS/axiodilaton critical points, not fully supersymmetric four-dimensional Minkowski vacua. For example $m_{\min}^2/m_{3/2}^2 = (28-12\sqrt{5})/10$ is independent of the common volume factor.

The exact periods avoid a large-complex-structure truncation in this model. The $g_s = 1/9$ construction improves the coupling relative to the earlier $S = 2\mathrm{i}$ anchor, but establishes neither corrected Kähler stabilization nor a localized warped ten-dimensional solution. Open-string fields, loop and alpha-prime corrections are not stabilized or bounded here. The reference to inherited geometric capacity 28 is conditional; no new geometric audit, global coupling optimum or universal tadpole statement is supplied.

The packet command `python reproduce.py` runs without network or old project paths. It verifies the symbolic two-integer and weak-coupling families, all 33 divisor assignments, the nine reduced point types, their 72 primitive pair costs, and a nontrivial modular image counted as the same orbit. It recomputes the signs, charge normalization, full real Jacobians and canonical spectra. Source hashes identify earlier completed results; the new portable output is distinguished from those inherited records.

A Seed calculation

In the interleaved ordering defined above, the real seed Jacobian is

$$J_{F,0} = \begin{pmatrix} -1 & -3 & 1 & -1 & 1 & -1 & 1 & -1 \\ -3 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & -3 & 1 & 1 & 1 & 1 \\ 1 & 1 & -3 & 1 & -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 & -1 & -3 & 1 & 1 \\ 1 & 1 & -1 & 1 & -3 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 & 1 & 1 & -1 & -3 \\ 1 & 1 & -1 & 1 & -1 & 1 & -3 & 1 \end{pmatrix}. \tag{25}$$

It follows from $\partial_{x_B} F_A = W_{AB} + \mathrm{i}W_B/(2y_A)$ and $\partial_{y_B} F_A = \mathrm{i}\partial_{x_B} F_A - \delta_{AB}\mathrm{i}W/(2y_A^2)$. At $y_A = 1$, (10) gives $H_0 = J_{F,0}^T J_{F,0}$. The script records both matrices and the factored characteristic polynomial.

Chronology and attribution disclosure

Earlier work was Fable 5-led; July 2026 onward results were predominantly produced by GPT models, as reported by the operator. DeepSeek and multiple GPT models also contributed cross-review and manuscript preparation. This is reported chronology, not a per-result authorship allocation. This note and its closure calculation are GPT-produced. Human author and affiliation fields are blank, and human expert review has not occurred. Historical source dates, the present delivery date of 29 October 2026, and demonstrable public disclosure are distinct; a private commit does not establish public priority.

> "I genuinely do not know how authorship, intellectual credit, or ownership should be attributed for these results."

References

[1] Steven B. Giddings, Shamit Kachru, and Joseph Polchinski. Hierarchies from fluxes in string compactifications. *Physical Review D*, 66:106006, 2002. doi: 10.1103/PhysRevD.66.106006. URL https://arxiv.org/abs/hep-th/0105097v2.

[2] Frederik Denef, Michael R. Douglas, Bogdan Florea, Antonella Grassi, and Shamit Kachru. Fixing all moduli in a simple F-Theory compactification. *Advances in Theoretical and Mathematical Physics*, 9:861–929, 2005. URL https://arxiv.org/abs/hep-th/0503124v2.

[3] Iosif Bena, Johan Blåbäck, Mariana Graña, and Severin Lüst. The tadpole problem. *Journal of High Energy Physics*, 2021(11):223, 2021. doi: 10.1007/JHEP11(2021)223. URL https://arxiv.org/abs/2010.10519v3.